

Heeseung Choi

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Research Interests

My research interests include, but are not limited to:

- RF/Microwave Circuits & Front-End Systems
- Applied Electromagnetics
- Wireless Communication

Education

Yonsei University, Seoul, South Korea

03/2020 - 02/2026

Bachelor of Engineering in Electrical and Electronic Engineering

GPA: 4.07/4.30

University of Georgia, GA, United States

08/2024 - 12/2024

Exchange Program

GPA: 4.00/4.00

Research Experience

Undergraduate Researcher

Seoul, South Korea

Yonsei University — Advisor: Prof. Byung-Wook Min

01/2025 – Present

- Led an undergraduate capstone on a 1-bit Phase Shifting Switch Module for Reconfigurable Intelligent Surface, optimizing transistor sizing and matching networks to achieve low insertion loss and a stable 180° phase shift over a wide bandwidth.
- Awarded the IEEE AP-S Undergraduate Research Scholarship; presented at KIEES (2025).
- Now advancing from 1-bit to 1.5-/2-bit switch modules, with prototypes entering fabrication, to cut quantization loss and improve beam control.

Publications & Presentations

- [1] **Heeseung Choi** and Byung-Wook Min, “1-bit Phase-Shifting Switch Module for Reconfigurable Intelligent Surfaces,” *KIEES Summer Conference*, Pyeongchang, South Korea, Aug. 2025 (Poster Presentation).

Awards & Scholarships

- **IEEE AP-S Undergraduate Summer Research Scholarship**
IEEE Antennas and Propagation Society
- **Korea-U.S. Technology Fields Exchange Program Scholarship**
Korea Institute for Advancement of Technology
- **Yonsei University Semiconductor Track Program**
Yonsei University
- **High Honors (Dean's List)**: Spring 2020, Spring & Fall 2023, Spring 2024, Spring 2025
Yonsei University

Skills

- **Tools**: Keysight ADS, HFSS, MATLAB, PSPICE/LTSPICE, VIVADO
- **Relevant Coursework**:
 - **Electromagnetism Track**: Electricity & Magnetism I/II, Microwave Engineering, Microwave Experiment, Antenna Engineering
 - **Communication Track**: Communication Theory, Communication Systems, Digital Signal Processing, Signals & Systems
 - **Microelectronics Track**: Electronic Circuits I/II, Physical Electronics